Genetic counselling

9A Genetics and evolution · Lesson 5 of 7

Date _____ Class _____

Today I can

- ☐ I can describe what a genetic test looks at and why babies and families may be tested.
- ☐ I can use probability and my knowledge of genes and chromosomes to explain a genetic test result simply, as a genetic counsellor would.

What happens today

1. Do Now: spot the difference
2. Learn: Reading a gene: sequences and faulty proteins · Genetic tests · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 14 to 15 · Workbook pages 10

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
genetic test	a test that looks at a person's DNA sequence or chromosomes
genetic counsellor	a person trained to explain genetic test results and the chances of passing on a condition
genetic disorder	a condition caused by a faulty gene or an unusual set of chromosomes
DNA sequence	the order of the four letters (A, T, C, G) along a DNA molecule
faulty gene	a gene whose sequence has changed so that it no longer works properly
haemoglobin	the protein in red blood cells that carries oxygen
sickle cell disease	a genetic disorder in which faulty haemoglobin makes red blood cells sickle-shaped
cystic fibrosis	a genetic disorder in which thick mucus builds up in the lungs and gut
Down's syndrome	a condition caused by an extra copy of chromosome 21
probability	the chance of something happening, e.g. 25% or 1/4
chromosome	a structure in the nucleus made of one long DNA molecule

Quick Review

spot the difference

DNA stores information as a code of four letters: A, T, C and G. Two people had the same gene read: P: CTT CTC GGA TCG CGG GAA ATA TTC R: CTT CTC GGA TCG CCG GAA ATA TTC

1 Find the difference. 2. What could a change in a gene do to a person? 3. Recap: what is a gene?

Core 1 of 3

Everyone does these. Answer on the card or under it.

1 People P, Q, R and S had the same gene analysed. Circle the group of letters that is different, and state which person has a gene that is not the same as the others.

P CTT CTC GGA TCG CGG GAA ATA TTC

Q CTT CTC GGA TCG CGG GAA ATA TTC

R CTT CTC GGA TCG CCG GAA ATA TTC

S CTT CTC GGA TCG CGG GAA ATA TTC

The person with the different gene is _____ .

2 Match each term to its meaning. Draw a line.

genetic test	a condition caused by a faulty gene or unusual chromosomes
genetic counsellor	the chance of something happening
genetic disorder	a test that looks at a person's DNA or chromosomes
probability	a person who explains genetic test results and chances to families

Core 2 of 3

Everyone does these. Answer on the card or under it.

3 The different version of the gene in question 1 causes a disorder in which waste substances cannot be removed from the body. Explain why finding out that a baby has this version of the gene is better than waiting until the baby is a child.

4 A test report reads: 'Both parents have a faulty version of the CFTR gene. There is a 25% chance of them having a child with cystic fibrosis.'

a) If the parents have four children, how many are predicted to have cystic fibrosis?

Core 3 of 3

Everyone does these. Answer on the card or under it.

4 *continued*

b) Explain why the real number could be different from your prediction.

Challenge 1 of 2

When Core is finished. Full sentences.

C1 A person's chromosome test shows an unusual arrangement of two chromosomes. The person has no health problems. A woman with chromosomes like this has a 10% chance of having a baby with Down's syndrome; a man with the same chromosomes has no increased risk. You are the genetic counsellor. The person tells you they want to start a family. Write what you would say. Include the probability, what it means, and what the options are.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 Suggest why genetic tests on babies are becoming more common, and give one argument against testing every baby for every known disorder.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What does a genetic test look at?

2 Both parents carry a faulty gene and each child has a 25% chance of the disorder. Write 25% as a fraction and as a decimal.

3 Give one thing a genetic counsellor does.

Today I can: Goal 1 ☐ Goal 2 ☐